ROGER HO

Pittsburgh, PA 15206 | rogerho@andrew.cmu.edu | +1 (734) 968-2525 | rogerksho.github.io

EDUCATION

CARNEGIE MELLON UNIVERSITY **Expected 2029**
PhD, Mechanical Engineering **Pittsburgh, PA**
Surgical Mechatronics Lab, Robotics Institute (Advisor: Cameron N. Riviere)
GPA: 4.00/4.00

UNIVERSITY OF MICHIGAN **Aug 2019 - May 2023**
BSE Mechanical Engineering; Minor in Computer Science **Ann Arbor, MI**
Overall GPA: 3.92/4.00, summa cum laude | **Major GPA:** 4.00/4.00

RESEARCH EXPERIENCE

SURGICAL MECHATRONICS LAB **Aug 2025 - Present**
Research Assistant | Carnegie Mellon University **Pittsburgh, PA**

- Developed a 6-DoF parallel kinematic handheld robotic micromanipulator, carrying the project from concept through design, fabrication, and testing, achieving superior precision and positioning bandwidth over previous iterations.
- Optimized design parameters using gradient-free particle swarm optimization to select design parameters that maximize kinematic isotropy over the workspace.
- Built a modular ROS 2 research platform to replace the legacy LabVIEW system, spanning real-time PC setup and C++/Python control nodes, enabling rapid prototyping of new controllers.
- Researching control methods that enforce safety and provide guidance under shared autonomy, such as control barrier function-based quadratic programming with calibrated safety guarantees.

PRECISION SYSTEMS DESIGN LABORATORY **May 2022 - May 2023**
Research Assistant | University of Michigan **Ann Arbor, MI**

- Devised and conducted experiments to characterize a novel flexure mechanism to verify the performance of a theoretical structural augmentation, resulting in a publication.
- Collaborated with a team to design a flexure-based parallel XY nanopositioning system with a focus on manufacturability, and carried the project through design, manufacturing, assembly, and open-loop testing, resulting in a patent.
- Carried out thorough static and dynamic finite element analysis (FEA) to optimize flexure parameters for actuation stiffnesses, bearing stiffnesses, and resonant frequencies.
- Developed a computational framework to simulate and subsequently select the appropriate precision actuators and sensors based on flexure parameters and desired motion profile.

WORK EXPERIENCE

STEALTH-MODE ROBOTICS STARTUP **Jul 2023 - Present**

Design Engineer **Ann Arbor, MI**

- Led the development of a robust mechatronic system for a robotic device, successfully demonstrating the function and novelty of our product to prospective users.
- Leading the development of miniature manipulators for a robotic device, meeting crucial company milestones in a fast-paced environment with stringent deadlines.
- Managed projects for multiple initiatives, entailing responsibilities such as risk management, mechanical design, mechanism analysis, testing and validation, design for manufacturing, and coordination of manufacturing logistics, while ensuring adherence to industry standards.

PUBLICATIONS

1. **Roger Ho**, Robert A. MacLachlan, Cameron N. Riviere "Active Constraints for Simultaneous Guidance and Safety in Handheld Micromanipulation" *under review*
2. Moeen Radgolchin, David Radtke, Siddharth Rath, **Roger Ho**, and Shorya Awtar, "Experimental characterization of a sandwich double parallelogram flexure mechanism," *Precision Engineering*, vol. 99, pp. 643–657, May 2026, doi: 10.1016/j.precisioneng.2026.02.002.

INVENTIONS

1. Shorya Awtar, **Roger Ho**, and Siddharth Rath, "Method of manufacture and assembly of XY flexure mechanism assembly," U.S. Patent 12,258,263 B1, Mar. 25, 2025.

HONORS

JAMES B. ANGELL SCHOLAR **Mar 2021 - Apr 2023**

Awarded to students who achieve all A's (A+, A, A-) for two or more consecutive terms.

UNIVERSITY HONORS/DEAN'S LIST **Dec 2019 - Apr 2023**

Awarded to students who earn a GPA of 3.5 or above during a term.

SOFTWARE EXPERIENCE

- **Languages:** MATLAB (Simulink, Simscape), LabVIEW, Python (Pytorch), Java, Javascript, SQL, C++, C, Verilog (Quartus II)
- **Mechanical:** Solidworks (CSWA Certified), ANSYS (Mechanical, Discovery), MSC Adams
- **Electronics:** LTSpice, KiCAD, Altium
- **Frameworks:** LabVIEW Real-Time, ROS2, Real-time Linux (PREEMPT_RT)